



IMMS

SMART TRACKING,
SAFE LIVING

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Introduction

Implanted microchips are small, versatile devices that are set under the skin to store and transmit data. They are gradually becoming more embedded in our daily lives, offering a blend of convenience, security and health benefits. This system aims to managing and monitoring microchips implanted in individuals (humans and animals).

To use IMMS, the user must register and activate the microchip through its unique ID. After registering, the user can manage various services such as:

- Managing Medical Data: storing and retrieving reports of their vital signs such as:
 - 1- Temperature
 - 2- Blood pressure
 - 3- Heart Rate
- Real-time Location Tracking: offering a real-time and accurate location.

Vision Statement

To create a silent and unseen guardian, an eternal watchful eye that never sleeps.

Purpose Statement

To enable individuals to monitor their health, enhance personal safety and improve quality of life.

Our Values

- Integrity: Honesty and trust in everything we do.
- Innovation: Encouraging creativity to lead the future.
- Responsibility: Taking ownership of our actions and decisions.
- Corporation: Prompting effective teamwork.
- User privacy: Protecting users' privacy with the highest standards of security.

Mission Statement

Offering an advanced, user-friendly and effective system for managing implanted microchips, by using up-to-date technologies.

Goals

- Promoting the concept of microchips.
- Defect-free system.
- Enhance accessibility through inclusivity.
- Reducing manufacturing costs.

Objectives

- Increasing awareness of individuals through consultations, seminars and workshops.
- Designing a well-structured system and using the latest technologies.
- Developing a user-friendly interface and ensuring usability for all individuals.
- Improving manufacturing efficiency through recycling and reuse.

SWOT Analysis



STRENGTH

- Good technical skills in software and hardware.
- Teamwork and communication.
- Strong motivation.

- Small team size.
- Limited budget.
- Little experience in the medical field.



WEAKNESSES



OPPORTUNITIES

- Unique idea.
- Working with research centers.
- Possibility for government or healthcare contracts.
- Capability of integrating the system with other intelligent systems (Hospitals systems).

- Fast changes in tech.
- Misunderstanding by the public or fear regarding implants.
- Government rules.



THREATS

Business Results

- Increased awareness among individuals about the benefits of microchips.
- Increased demand for microchips.
- Expanded the customer base by 30%, boosting market share by 10%.

Why are we doing this project?

- With the world witnessing great technological advances and artificial intelligence, Innovation has become a necessity.
- We decided to participate in this challenge to come up with an idea that hasn't yet been adopted in the tech industry.
- The aim of this idea is to minimize the time wasted by humans, get more efficiency and access to real-time services.
- Allowing IMMS to provide more reliable services, especially in emergencies or high security situations.

Project Scope (MoSCoW)

- Storing personal and health data in a secure database.
- Integration between hardware and software components.
- Ability to add, update, and view health records.
- Set up real-time monitoring and alerts when there is something abnormal.

Mo
(Must-have)

S
(Should-have)

- Exportable health reports (PDF or Excel).
- Graphs to visualize changes in vital signs over time.
- Notification system via email or SMS.
- Real-time GPS tracking.
- Mobile app for real-time monitoring.

- Integration with other smart devices (smart watch).
- Connecting multiple microchips to a single account.
- Multi-language support.
- Calculate the number of steps and calories burned.

C
(Could-have)

W
(Won't-have)
At this point

- Cloud integration.
- Calculate blood sugar level.
- Integration with other fitness platforms.
- Messages between end-users.

Feasibility Study

1- Technical Feasibility:

The system will depend on two types of microchips: one implantable under the skin and another attachable to the skin.

Necessary components include:

- Hardware:
 - Office computers / Network components.
 - Microchips with built-in sensors.
 - Wireless transmitter (Bluetooth or RFID).
 - Long-life battery units.
 - Server (for backend data storage and processing).
- Software:
 - Mobile application for users to view data using Flutter & Dart.
 - Backend system for processing using Java.
 - Storage system for storing data using MySQL.
- Human resources needed:
 - Hardware engineers (to design and test the chip).
 - Embedded systems developer.
 - Software developers (for development).
 - UI/UX designers (for user-friendly interfaces).
 - Mobile developers.
 - Testers.

2- Operational Feasibility:

- **Management Support:** The project is fully supported by management due to its innovation and potential impact on healthcare.
- **User Support:** Users show strong interest and support for the system.
- **Current System Evaluation:** Existing systems are limited and not sufficient for continuous health tracking.
- **Need for Change:** Users recognize the need for a more advanced, automated system.
- **Workforce Impact:** The system is not intended to reduce workforce but rather to complement and support healthcare workers.
- **User Training Requirement:** A basic training will be required to help users and staff understand the system and its operations.
- **Training Resources:** The company is ready to provide support materials such as guides, tutorials, and support to ensure a smooth transition.

3- Schedule Feasibility:

Project Duration (32 Weeks):

- Project Planning & Analysis (Week 1 To Week 4).
- Systems Design (Week 5 To Week 8).
- Hardware & Software Development (Week 9 To Week 24).
- Integration & Testing (Week 25 To Week 28).
- Training Staff (Week 29 To Week 30).
- Pilot Launch (Week 31 To Week 32).

4- Economic Feasibility:

Breaking economic feasibility into 3 parts:

- Hardware:

	Attachable chip	Implanted Chip
Base cost	50\$	125\$
Selling price	149.99\$	299.99\$

- Office computers (20 units): 12,000\$
 - Servers (2 units): 5,000\$
 - Network components: 3,000\$
 - Other accessories: 2,500\$
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- Software:
 - Software development: 24,000\$
 - Microsoft SQL server standard (database): 2,500\$
 - Operating system (Linux): Free
 - Security software (antivirus, firewall): 4,000\$
 - Software testing: 16,000\$
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- Employees' Salaries (annually):
 - Hardware engineers: 75,000\$ – 85,000\$
 - Software engineers: 80,000\$ - 95,000\$
 - Embedded systems developers: 70,000 – 80,000\$
 - UI/UX Designers: 70,000\$ - 80,000\$
 - Testers: 70,000\$ - 90,000\$

Gantt Chart

